A Nonlinear Two-Path Regression Test in Factor Asset Pricing Models

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Contents

1 Empirical Studies **1**

1.1 Test Procedure 2

1.2 Factor Zoo Individual Test 3

2 A second method **9**

1 Empirical Studies

Let Y_{it} be the excess returns, F_t be the k-dim factors. We assume the market model:

$$Y_{it} = a_i + b_i' F_t + \varepsilon_{it} \tag{1}$$

Fiorentini, Galesi, and Sentana, 2018 Many economic theories imply that the expected return is a linear function of betas:

$$E(Y_i \mid b_i) = b_i' \gamma \tag{2}$$

and we test this relationship.

We have two sets of data:

1. Factor zoo data and monthly return data for all CRSP stocks

(a) Factor zoo data from Feng, Giglio, and Xiu, 2017, 1976-06 to 2017-12, monthly data of 147 factors.

(b) Monthly holding-period return data of 463 stocks on CRSP that don't have NA in the sample period. The total number of available stocks are 29,742, but (too) many have NA.

2. CRSP Beta Suite dataset:

   For SP500 constituents from 1990-01-02 to 2019-12-31, we have daily data on

   (a) return;

   (b) betas on Carhart's four factors, calculated using moving average, so the betas have variation.

   (c) idiosyncratic and total volatility.

   The daily risk-free rate is taken from French's website. After dropping NA, we are left with 253 stocks, but it can be expanded to incorporate more stocks.

1.1 Test Procedure

1. For each $i = 1, \ldots, n$, I did the following regression, let Y_{it} be the excess return $R_{it} - R_t^f$, let F_t be k-dim factor excess return

$$Y_{it} = a_i + b_i' F_t + \varepsilon_{it} \tag{3}$$

   obtain estimates $\hat{a}_i, \hat{b}_i$;

2. The cross-sectional regression of $\bar{Y}_i = \frac{1}{T}\sum_t Y_{it}$ on $\hat{b}_i$. If we include the intercept, then we are only test that the expected return is a linear function of betas, i.e., there exist $(\alpha, \gamma) \in \mathbb{R} \times \mathbb{R}^k$ such that

$$E(Y_i \mid b_i) = \alpha + b_i'\gamma \tag{4}$$

   whereas if we do not include the intercept in this step, we are testing for both

linearity and no significant pricing error, there exists γ:

$$E(Y_i \mid b_i) = b_i'\gamma \tag{5}$$

In either case, we obtain the estimate $\hat{\gamma}$ and the residual $\hat{u}_i = \bar{Y}_i - \hat{b}_i'\hat{\gamma}$.

3. Collect $(\hat{u}_i, i = 1, \ldots, n)$ and construct a specification test statistic as in Li and Racine, 2007, which is very close to Hardle and Mammen, 1993. First we calculate

$$I_n := \frac{1}{n(n-1)} \sum \hat{u}_i \hat{u}_j K_{h,ij} \tag{6}$$

where $\hat{K}_{h,ij} = \prod_{s=1}^{k} h_s^{-1} K\left(\frac{\hat{b}_{is} - \hat{b}_{js}}{h_s}\right)$, I used Guassian density $K(\cdot)$ and the bandwidth for the s-th beta is chosen to be $h_s = 1.06 \times \text{std}(\hat{b_s}) \times n^{-1/5}$ for simplicity. I_n corresponds to the hypothesis:

$$H_0 : E(uE(u \mid b)f(b)) = 0, a.s. - b \tag{7}$$

The test statistic is

$$T_n = \frac{n(h_1 \ldots h_k)^{\frac{1}{2}}}{\sqrt{\hat{\sigma}^2}} I_n \tag{8}$$

where $\hat{\sigma}^2 = \frac{2(h_1 \ldots h_k)}{n(n-1)} \sum_{i \neq j}^{n} \hat{u}_i^2 \hat{u}_j^2 \hat{K}_{h,ij}^2$

4. Use wild bootstrap to approximate the distribution of T_n.

1.2 Factor Zoo Individual Test

With the first set of data, we did the following tests, with Factor X taking each factor in the zoo:

1. Test each factor individually, that is we assume there is only one Factor X.
2. Test Market Factor + Factor X.
3. Test FF3 Factors + Factor X.

In the second-step cross sectional regression, we have incorporated an intercept, so the test is purely for nonlinearity.

	Factor X	MktRf + Factor X	FF3 + Factor X
Number of Rejections	34	30	41

Table 1: Number Of Rejections for Different Models

Question 1. I think the reason model *FF3* + *X* has more rejections is because if we use full sample, then MktRf and HML are both rejected for being linear. I wonder if it makes more sense to split the sample.

The following is the table of result. The numbers are 1. test statistic value and bootstrap distribution quantiles. Red ones are those rejected.

Name of Factor X	Factor X	MktRf + Factor X	FF3 + Factor X
	(T_n, Bootstrap 2.5% and 97.5% quantile)		
BAB	(0.687,-1.443,1.636)	(2.975,-1.622,1.803)	(2.820,-1.778,2.216)
CMA	(-0.858,-1.400,1.083)	(0.516,-1.549,1.532)	(0.274,-1.797,1.777)
HML	(2.126,-1.453,1.458)	(0.014,-1.742,1.579)	(—,—,—)
HML_Devil	(2.230,-1.461,1.918)	(0.812,-1.644,1.613)	(2.006,-1.884,1.709)
HXZ_IA	(-0.134,-1.429,1.133)	(0.498,-1.556,1.317)	(0.767,-1.922,1.266)
HXZ_ROE	(0.633,-1.389,1.552)	(2.659,-1.641,1.539)	(1.732,-1.835,2.028)
IPO	(-0.757,-1.463,1.224)	(2.338,-1.767,1.651)	(2.450,-1.871,2.175)
LIQ_PS	(3.033,-1.385,1.360)	(1.878,-1.635,2.289)	(0.725,-1.729,2.027)
LTR	(-0.952,-1.463,1.099)	(1.522,-1.698,1.550)	(1.310,-1.865,1.561)
MktRf	(2.144,-1.450,1.415)	(—,—,—)	(—,—,—)
QMJ	(-0.778,-1.478,1.358)	(1.513,-1.736,2.028)	(1.095,-1.760,1.789)
RF	(2.462,-1.420,0.964)	(1.671,-1.760,2.044)	(1.376,-1.733,1.657)
RMW	(0.203,-1.421,1.132)	(0.397,-1.737,1.420)	(2.693,-1.689,1.705)
SMB	(-0.185,-1.494,1.267)	(0.510,-1.768,1.706)	(—,—,—)
STR	(1.219,-1.480,1.274)	(3.128,-1.830,1.456)	(0.520,-1.906,1.571)
UMD	(0.332,-1.384,1.131)	(0.018,-1.706,1.611)	(1.071,-1.843,1.997)
absacc	(1.299,-1.324,1.460)	(0.956,-1.617,2.012)	(2.351,-1.847,1.576)
acc	(7.448,-1.469,2.029)	(2.096,-1.720,1.597)	(1.682,-1.928,1.941)
adm	(1.192,-1.237,1.564)	(1.689,-1.514,1.599)	(1.229,-1.886,1.805)
aeavol	(-0.755,-1.326,1.199)	(-0.238,-1.502,1.712)	(0.127,-1.795,2.034)
age	(-0.684,-1.379,1.255)	(0.494,-1.654,1.638)	(2.300,-1.860,1.451)
ala	(-0.330,-1.482,1.741)	(0.737,-1.646,1.712)	(1.523,-1.680,2.094)
ato	(-0.837,-1.344,1.442)	(2.233,-1.774,1.864)	(1.945,-1.870,1.560)
baspread	(0.475,-1.347,1.156)	(1.510,-1.618,1.806)	(3.327,-1.776,1.746)
beta	(0.554,-1.375,1.392)	(2.441,-1.608,1.810)	(3.255,-1.655,1.787)
bm_ia	(0.673,-1.352,1.548)	(0.990,-1.716,1.653)	(1.136,-1.881,1.435)
cash	(-0.143,-1.303,0.909)	(0.376,-1.592,1.635)	(2.619,-1.941,2.069)
cashdebt	(0.233,-1.455,1.292)	(1.632,-1.747,1.658)	(-0.097,-1.781,1.749)
cashpr	(-0.598,-1.417,1.159)	(-0.880,-1.597,1.740)	(0.639,-1.732,1.629)
cdi	(0.375,-1.445,1.182)	(2.358,-1.640,1.726)	(1.314,-1.880,1.684)
cei	(-0.403,-1.432,1.751)	(0.644,-1.605,1.755)	(0.224,-1.711,1.926)
cfp	(-0.774,-1.377,1.036)	(1.273,-1.716,1.951)	(1.734,-2.068,1.683)
cfp_ia	(-0.953,-1.447,1.313)	(-0.245,-1.654,1.748)	(1.015,-1.760,1.673)
chatoia	(0.067,-1.462,1.387)	(1.212,-1.680,1.925)	(0.365,-1.946,1.775)
chcsho	(-0.567,-1.374,1.206)	(1.130,-1.472,1.453)	(2.857,-1.723,1.878)
chempia	(-0.113,-1.499,1.120)	(0.927,-1.528,1.752)	(0.967,-1.843,1.978)
chinv	(-0.032,-1.434,0.949)	(0.360,-1.615,1.700)	(1.068,-1.785,2.108)
chmom	(-0.538,-1.493,1.230)	(1.670,-1.647,1.743)	(0.916,-1.814,2.102)
chpmia	(4.210,-1.468,1.144)	(1.422,-1.738,1.931)	(0.614,-1.853,1.624)
chtx	(1.741,-1.467,1.292)	(-0.072,-1.823,1.612)	(0.906,-1.908,1.710)
cinvest	(-0.713,-1.381,1.284)	(0.057,-1.743,1.936)	(-0.576,-2.055,1.734)

cinvest_a	(0.423,-1.360,1.309)	(0.732,-1.687,1.503)	(1.222,-2.057,1.535)
convind	(-0.218,-1.373,1.201)	(0.379,-1.643,1.473)	(1.394,-1.965,2.196)
cp	(-1.133,-1.326,0.972)	(0.468,-1.798,1.718)	(0.490,-1.929,1.694)
cto	(-0.693,-1.467,1.048)	(1.386,-1.721,1.611)	(0.528,-1.815,1.962)
currat	(-0.881,-1.359,1.024)	(1.019,-1.487,1.552)	(2.060,-1.742,1.846)
dcoa	(-0.863,-1.450,1.182)	(-0.243,-1.718,1.655)	(0.135,-1.789,1.784)
dcol	(-0.454,-1.387,0.950)	(1.528,-1.517,1.534)	(1.689,-2.040,1.581)
depr	(-0.659,-1.338,1.138)	(0.288,-1.559,1.572)	(2.824,-1.866,2.116)
dfin	(-0.149,-1.268,1.575)	(1.058,-1.779,1.311)	(1.141,-1.949,1.634)
dfnl	(1.797,-1.460,1.570)	(0.484,-1.841,1.907)	(0.564,-2.142,1.931)
dnca	(3.985,-1.330,1.159)	(1.264,-1.678,1.752)	(0.588,-1.710,1.663)
dncl	(-0.603,-1.386,1.525)	(1.438,-1.661,1.605)	(1.150,-2.074,1.576)
dnco	(-1.272,-1.373,1.426)	(0.749,-1.564,1.382)	(1.294,-1.816,1.684)
dnoa	(1.930,-1.478,1.434)	(0.727,-1.657,1.607)	(0.646,-1.900,1.606)
dolvol	(3.982,-1.396,1.164)	(3.947,-1.795,2.157)	(3.187,-1.739,1.937)
dpia	(1.287,-1.367,1.425)	(0.377,-1.524,1.841)	(1.372,-1.829,1.683)
dsti	(6.661,-1.385,1.647)	(1.338,-1.705,1.759)	(2.237,-1.908,2.245)
dwc	(-0.919,-1.443,1.430)	(0.044,-1.610,1.838)	(1.246,-1.831,1.957)
dy	(-0.831,-1.303,1.263)	(0.802,-1.647,1.777)	(2.258,-1.942,1.637)
ear	(-0.148,-1.448,1.431)	(2.553,-1.788,2.261)	(1.711,-2.020,2.102)
ebp	(0.439,-1.279,1.365)	(-0.127,-1.529,1.339)	(0.869,-1.866,1.698)
egr	(0.078,-1.424,1.134)	(0.399,-1.703,1.278)	(-0.205,-1.918,1.899)
egr_hxz	(-0.415,-1.345,0.931)	(-0.274,-1.660,1.263)	(0.424,-1.853,1.686)
em	(2.163,-1.469,1.016)	(0.920,-1.537,1.297)	(-0.034,-1.757,1.971)
ep	(-0.551,-1.281,1.117)	(0.172,-1.743,1.719)	(3.853,-1.940,1.830)
etr	(1.185,-1.404,1.074)	(0.285,-1.856,1.803)	(0.392,-2.021,1.665)
gad	(2.053,-1.260,1.145)	(0.524,-1.488,1.655)	(0.532,-1.935,1.922)
gma	(0.208,-1.427,1.379)	(0.424,-1.821,1.622)	(0.439,-1.831,1.936)
grcapx	(0.825,-1.459,1.184)	(1.793,-1.749,1.929)	(1.477,-1.874,1.572)
grltnoa	(1.713,-1.427,1.584)	(0.890,-1.468,1.381)	(1.514,-1.865,2.005)
grltnoa_hxz	(2.589,-1.449,1.416)	(1.003,-1.791,1.873)	(0.426,-1.874,1.835)
herf	(1.008,-1.431,0.834)	(3.403,-1.493,1.410)	(2.541,-1.831,1.421)
hire	(-0.735,-1.412,1.041)	(1.083,-1.775,1.208)	(0.485,-1.851,1.869)
idiovol	(1.138,-1.429,1.936)	(1.907,-1.763,1.583)	(2.695,-1.827,1.842)
ill	(1.892,-1.477,1.946)	(0.473,-1.596,1.893)	(0.130,-1.871,1.705)
indmom	(-0.259,-1.378,1.734)	(0.913,-1.528,1.367)	(-0.125,-1.903,1.320)
invest	(1.627,-1.353,1.298)	(0.374,-1.528,1.845)	(1.400,-1.812,1.772)
ivg	(2.508,-1.454,1.167)	(1.442,-1.837,1.640)	(0.760,-1.811,1.978)

kz	(0.467,-1.354,1.887)	(-0.350,-1.587,1.413)	(0.470,-1.724,1.447)
lev	(0.019,-1.366,1.461)	(-0.224,-1.628,1.658)	(2.379,-2.007,1.289)
lfe	(-0.053,-1.397,0.777)	(1.038,-1.639,1.780)	(-0.051,-1.795,1.780)
lgr	(-0.713,-1.425,1.269)	(1.773,-1.653,1.504)	(0.342,-1.869,2.183)
maxret	(0.093,-1.372,1.266)	(0.846,-1.890,1.537)	(2.429,-1.806,1.711)
mom36m	(1.847,-1.475,1.349)	(1.609,-1.816,1.573)	(0.730,-1.775,2.041)
mom6m	(0.602,-1.499,1.556)	(0.998,-1.579,1.689)	(0.315,-1.944,1.698)
moms12m	(2.055,-1.422,1.301)	(0.226,-1.669,1.406)	(0.512,-1.736,1.689)
ms	(-1.084,-1.470,1.506)	(0.621,-1.714,1.997)	(0.607,-1.803,1.864)
ndf	(2.431,-1.455,1.218)	(1.540,-1.934,1.528)	(0.188,-1.759,1.732)
ndp	(-0.875,-1.273,1.070)	(0.658,-1.649,1.560)	(1.864,-1.997,1.433)
nef	(-0.484,-1.425,1.854)	(2.485,-1.701,1.819)	(1.688,-1.779,1.794)
nincr	(-0.504,-1.328,0.950)	(0.656,-1.732,1.789)	(0.903,-1.862,1.541)
noa	(0.553,-1.334,1.533)	(0.485,-1.725,1.517)	(3.193,-1.984,2.069)
nop	(-0.455,-1.387,1.242)	(-0.188,-1.716,1.624)	(1.561,-1.654,1.760)
nxf	(-0.016,-1.428,1.352)	(0.991,-1.843,1.629)	(1.349,-1.717,1.691)
ob_a	(1.603,-1.495,1.444)	(0.231,-1.594,2.015)	(0.091,-1.803,1.660)
ol	(0.086,-1.449,1.515)	(1.779,-1.540,1.677)	(0.114,-1.723,1.867)
op	(-0.152,-1.328,1.382)	(0.315,-1.713,1.726)	(1.493,-1.673,1.661)
orgcap	(1.450,-1.435,1.491)	(0.656,-1.617,1.798)	(0.745,-1.663,1.797)
os	(-0.989,-1.526,1.655)	(1.369,-1.816,1.653)	(-0.872,-1.881,1.594)
pchcapx	(-0.450,-1.445,1.065)	(0.991,-1.691,1.507)	(0.607,-1.728,1.592)
pchcapx3	(-0.378,-1.426,1.274)	(1.269,-1.624,1.586)	(0.331,-1.799,1.575)
pchcapx_ia	(1.596,-1.448,1.496)	(2.784,-1.684,1.720)	(-0.103,-1.843,1.603)
pchcurrat	(4.353,-1.367,2.208)	(2.972,-1.669,2.077)	(2.672,-1.920,1.522)
pchdepr	(-0.793,-1.424,1.345)	(0.803,-1.607,1.474)	(2.019,-1.939,1.753)
pchgm_pchsale	(3.384,-1.384,1.866)	(1.614,-1.671,1.804)	(1.302,-1.677,2.320)
pchquick	(4.237,-1.468,1.700)	(2.500,-1.912,1.986)	(1.445,-1.935,2.296)
pchsale_pchinvt	(0.521,-1.531,1.597)	(1.397,-1.807,1.345)	(0.120,-1.969,1.465)
pchsale_pchrect	(1.579,-1.496,1.318)	(1.750,-1.882,1.303)	(2.090,-1.944,1.711)
pchsale_pchxsga	(0.891,-1.423,1.663)	(0.213,-1.817,1.429)	(0.479,-1.883,1.906)
pchsaleinv	(-0.371,-1.520,1.165)	(1.474,-1.794,1.679)	(-0.062,-1.999,1.669)
pctacc	(0.130,-1.506,1.387)	(-1.155,-1.924,1.603)	(1.266,-2.025,2.072)
pm	(-0.332,-1.486,1.877)	(0.425,-1.702,1.388)	(-0.126,-1.698,1.611)
poa	(0.028,-1.392,1.418)	(-0.399,-1.887,1.653)	(2.095,-1.673,2.133)
pps	(-0.554,-1.467,1.521)	(2.088,-1.699,1.493)	(1.316,-2.000,1.656)
pricedelay	(0.048,-1.438,1.155)	(1.231,-1.630,1.491)	(1.270,-1.676,1.757)
ps	(0.070,-1.402,1.013)	(1.720,-1.626,1.639)	(1.852,-1.923,2.399)
quick	(-0.454,-1.298,1.320)	(0.104,-1.660,1.626)	(2.042,-1.855,1.646)
rd	(1.235,-1.385,1.293)	(0.686,-1.605,1.393)	(2.045,-2.052,1.939)

rdm	(-0.668,-1.334,1.046)	(0.997,-1.653,1.659)	(1.644,-1.846,2.310)
rds	(-0.294,-1.336,1.278)	(1.032,-1.641,1.867)	(1.422,-1.871,2.003)
realestate_hxz	(4.818,-1.407,1.671)	(1.314,-1.704,2.302)	(1.895,-1.874,1.946)
retvol	(0.538,-1.374,1.410)	(1.947,-1.664,1.994)	(2.378,-1.646,1.813)
rna	(0.345,-1.416,1.486)	(2.723,-1.552,1.425)	(1.579,-1.873,1.738)
roaq	(-0.235,-1.423,1.477)	(0.760,-1.773,1.949)	(1.355,-1.780,1.933)
roavol	(-0.549,-1.298,1.444)	(0.403,-1.642,1.332)	(2.956,-1.891,1.838)
roic	(-0.346,-1.526,1.441)	(0.438,-1.828,1.678)	(0.780,-1.938,1.941)
rs	(0.145,-1.444,1.043)	(0.894,-1.679,1.380)	(1.696,-1.760,1.973)
rsup	(7.620,-1.660,1.791)	(3.774,-1.879,1.396)	(4.245,-1.892,1.760)
salecash	(-0.498,-1.295,0.905)	(0.618,-1.535,1.481)	(3.306,-1.794,2.261)
saleinv	(0.494,-1.353,1.405)	(2.000,-1.548,1.529)	(1.280,-1.839,2.006)
salerec	(1.576,-1.294,1.621)	(1.666,-1.604,1.792)	(0.488,-1.813,1.652)
sgr	(-0.569,-1.402,1.068)	(1.314,-1.628,1.923)	(-0.166,-1.852,1.961)
sin	(0.813,-1.396,1.686)	(1.965,-1.765,1.813)	(0.323,-1.829,1.820)
sp	(6.008,-1.436,2.234)	(0.889,-1.684,1.648)	(1.824,-1.710,1.576)
std_dolvol	(-0.574,-1.414,1.171)	(1.399,-1.740,2.066)	(0.797,-1.710,2.049)
std_turn	(-0.369,-1.318,1.343)	(0.613,-1.854,1.429)	(2.124,-1.827,1.973)
stdacc	(-0.192,-1.387,1.411)	(0.677,-1.696,1.703)	(1.687,-1.969,1.753)
stdcf	(0.166,-1.382,1.550)	(0.742,-1.651,1.527)	(1.588,-1.948,1.875)
sue	(0.730,-1.418,1.830)	(-1.173,-1.767,1.553)	(-0.879,-1.910,2.054)
ta	(1.982,-1.441,1.655)	(0.716,-1.696,1.552)	(0.912,-1.926,1.734)
tang	(0.015,-1.353,1.349)	(1.272,-1.619,1.896)	(2.629,-1.869,1.523)
tb	(-0.629,-1.390,1.370)	(2.196,-1.545,1.728)	(2.121,-1.693,1.844)
turn	(0.156,-1.331,1.214)	(0.650,-1.736,1.453)	(1.835,-1.791,1.782)
ww	(-0.587,-1.381,1.577)	(1.681,-1.692,1.802)	(2.706,-2.022,1.559)
zerotrade	(0.231,-1.380,1.258)	(0.843,-1.765,1.632)	(2.054,-1.786,1.808)
zs	(0.212,-1.323,1.744)	(-0.160,-1.568,1.545)	(2.027,-2.099,1.271)

2 A second method

Let $C_v = (E_1 - I_k v)$, let $\theta_i = (a_i, b_i')'$, then $a_i - b_i' v = C_v' \theta_i$. Let $\hat{u}_i = C_{\hat{v}}' \hat{\theta}_i$. We can construct a test based

$$\mathbf{T}_{N,T} = \frac{2}{n(n-1)} \sum_i \sum_{j \neq i} \hat{u}_i \hat{u}_j \hat{K}_{ij,h} \tag{9}$$

where $\hat{K}_{ij,h} = \frac{1}{h} K\left(\frac{\hat{X}_i - \hat{X}_j}{h}\right)$.

Test statistic is also

$$\frac{1}{n(n-1)} \hat{U}' \hat{\mathbf{K}} \hat{U} \tag{10}$$

$$\mathbf{T}_{N,T} = \frac{1}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} \tag{11}$$

We first provide an estimate for $\hat{u}_i$.

$$\begin{aligned}
\hat{u}_i &= C_{\hat{v}}' \left(\hat{\theta} - \theta\right) + (C_{\hat{v}}' - C_v')\theta && (12)\\
&= C_v' \left(\hat{\theta} - \theta\right) + (C_{\hat{v}}' - C_v')\left(\hat{\theta} - \theta\right) + (C_{\hat{v}}' - C_v')\theta && (13)\\
&= C_v' \left(\hat{\theta} - \theta\right) + error && (14)
\end{aligned}$$

Now consider the term $u_i := C_v' \left(\hat{\theta} - \theta\right)$:

$$u_i = C_v' (F'F)^{-1} F' \varepsilon_i = \frac{1}{T} \sum_t \nu_t \varepsilon_{it} \tag{15}$$

where ν_t is common to all i.

References

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